

Local public goods as amenities: Evidence from London

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Abstract

This paper seeks to shed light on the role of local public good provision on location sorting and city characteristics. I augment a classic quantitative urban economics model to integrate local public goods into the residential amenity. I then use detailed data on London and take advantage of a natural experiment to parametrise the model. The model is used to simulate a policy counterfactual in London, where the government is enacting policy to further increase the degree of fiscal decentralisation. I show that this has important distributional consequences driven by how workers and firms decide to sort. This results in higher equilibrium prices and wages in areas that benefit from decentralisation, combined with an outflow of firms and workers from areas that are hit harder.

Introduction

Rising inequality within cities has been a crucial talking point over the past few decades. While several factors have contributed to this rise, including steep housing costs and a widening income gap, this inequality has brought the issue of local policy back to the forefront of the debate. The past fifteen years have seen many European governments implement fiscal austerity, decreasing the number of local services. This has been combined with increasing levels of fiscal decentralisation, particularly in the UK. The net results is a strengthening of the **inequality of services across different locations**, which has so far been overlooked in the revived urban economics literature. **and of course you are going to show convincing evidence of this claim.**

A new class of empirical urban models has overtaken the field and has helped unravel and understand the intricacies of how people choose where they live, work, and build in a city (Ahlfeldt et al., 2015; Diamond & McQuade, 2019). In addition, some recent contributions have helped open the black box of residential amenities (Almagro & Dominguez-Iino, 2022). However, a key factor has until now been excluded, and that is the role of local government in defining and providing residential amenities. In fact, local governments, especially in London, **collect taxes and allocate resources to various services**, such as schools, streets, cleanliness, garbage collection, and libraries. **These are all elements that we know people consider when choosing where to live (Zahra et al., 2018).**

The aim of this paper is to incorporate the role of local governments in the context of an empirical urban model. I borrow from the literature to build and then augment a simple

urban model with the addition of local public goods. I take advantage of a recent natural experiment in London that strongly affected the budgets of local councils, and then apply a border discontinuity difference in difference design. This allows me to estimate an elasticity of house prices to per capita local public good spending, taken as a proxy of quality, of just above 8.8%. The approach is inspired by the literature on house prices and schooling (Black, 1999), and **confirms the fact that buyers internalise the value of local public goods**. The estimate in this paper is slightly higher than the estimates from the school zone literature, but it is justifiable considering this figure represents a comprehensive package of local public goods, and not just schools.

The next contribution of the paper is to estimate a gravity equation on commuting flows in London. I retrieve a coefficient of 4.4, in line with recent findings in the literature (Heblich et al., 2020; Monte et al., 2018). I can then integrate these two reduced form estimates into the augmented urban model and estimate the structural model fully. First, I estimate the Fréchet parameter, finding a value of 3.53, slightly below the findings in the literature (Ahlfeldt et al., 2015; Eaton & Kortum, 2002). I am also able to deduce wages which show a strong degree of correlation with the data. Next, I can estimate the floor space supply elasticity in London, finding a value of 0.82, which is in line with cities such as New York and San Francisco and suggests that London faces strong constraints to construction (Saiz, 2010). With these parameters, **I am able to back out estimates of land and floorspace prices, showing their concentration within the city center**. By inverting the model, I can back out amenity and productivity terms. Finally, I can estimate the density externality on productivity and show that it is in line with other recent findings (Tsivanidis, 2019).

The model will allow me to simulate counterfactual policies or relevance. The British government is currently planning on further decentralising local authority funding, which will have the effect of additionally increasing funding inequality between boroughs. Understanding the first and second order distributional consequences of such a policy crucial, especially given the context of concentrated deprivation in the already under-funded areas. The counterfactual thus enables me to assess how sorting, wages, prices, and inequality will develop in general equilibrium. From the model, I find a strong price effect of the policy. Floor space prices and rents increase in the **peripheral areas least hit by the policy** - this is driven by greater demand to live in these areas. It is clear that the distributional consequences of this policy are large and consistent, and merit further analysis in the context of fiscal decentralisation.

This paper seeks to contribute to several different strands of literature. First, I augment a model in the literature (Sturm et al., 2023), and update our understanding of the parameters for a developed and vast city such as London to enable comparability across countries (Saiz, 2010). Additionally, I seek to build on the recent developments of quantitative urban models (Ahlfeldt et al., 2015; Heblich et al., 2020; Diamond & McQuade, 2019) by offering a fresh perspective on amenities and the interaction with local government. I offer an important avenue for utilizing these class of models within the fiscal decentralisation discussion, stressing the importance of local public goods when discussing inequality (Diamond et al., 2019; Almagro et al., 2023). I further provide evidence that amenities are not just either fundamental or endogenous (Almagro & Dominguez-Iino, 2022), but rather that public goods are

amenities and that policy matters.

Finally, my paper will also fit within the fiscal migration literature. This literature goes all the way back to Tiebout (1956)’s theory of local expenditure. The focal point of his hypothesis is that local councils can compete for residents by providing different tax/policy packages, and that people would be free to choose the package that best suits them (i.e. “voting with your feet”). The literature developed further, with Davies (1982) augmenting the Aronson-Schwartz model to deal specifically with London’s particular fiscal structure. The underlying idea once again was that residents would move across the city in a way that was consistent with a “value-for-money” view of services provided by local councils. This preliminary model did in fact show a relationship between migration and fiscal attractiveness of councils. Whilst both of these papers provided key intuition on how residents sort, it is only by incorporating local government into a modern empirical urban model that we can fully appreciate these effects in general equilibrium. I aim to revive and contribute to this literature by doing so.

The rest of the paper is structured as follows. Section 1 explains the institutional context of the policy change, along with the functioning and funding of local government in the UK. Section 2 outlines all components of the model and the equilibrium. Section 3 will present all the empirical results of the paper, including the estimation of the model. Finally, in Section 4 I conduct the counterfactual simulation and in Section 5 I will discuss future steps for the project.

1 Institutional Setting

The study is set in London, whose urban economic setting stands unique in its intricate structure. London’s institutional framework begins with its boroughs, which are in full effect the equivalent to English local authorities (i.e. councils)¹, with slightly different powers to other local authorities in the country. The Greater London Authority (GLA) acts as an overarching distinct regional authority, operating strategically across London boroughs with separate powers and remit.

The economic importance and influence of boroughs in London cannot be overstated. They have exclusive authority over crucial services such as education, libraries, parks and recreation, council housing, waste collection and disposal, environmental health, and revenue collection. It is only with regards to larger strategic matters such as transport infrastructure, planning, and passenger transport that they share responsibility with the GLA. Finally, the GLA has exclusive authority over police and fire services.

Councils fund these activities via a mix of local tax collection and central government grants. The two main forms of local authority taxation are “council taxes”, collected on domestic properties, and “business rates”, collected on commercial properties. Prior to 2010, a large

¹Throughout the paper, I will use the terms borough, council and local authority interchangeably, consistent with the institutional framework of London

part of tax collected was handed back to the central government, which would redistribute funds to councils according to a specific formula.

Beginning in 2010, a large policy shift was undertaken by the newly elected British government. The aim was to reform the system of local government funding over the next few fiscal years, driven by a need to cut central government spending. By 2013, councils retained the entirety of council tax and were allowed to retain 50% of business rate collection. This was accompanied by large cuts in grants from the government to councils. In London, the result was an average reduction in council budgets per capita by nearly 20% across the city, as shown in Figure (1). The worst hit boroughs were generally the ones in inner London, with the evident exception of the City of London that largely gained via the introduction of business rate retention.

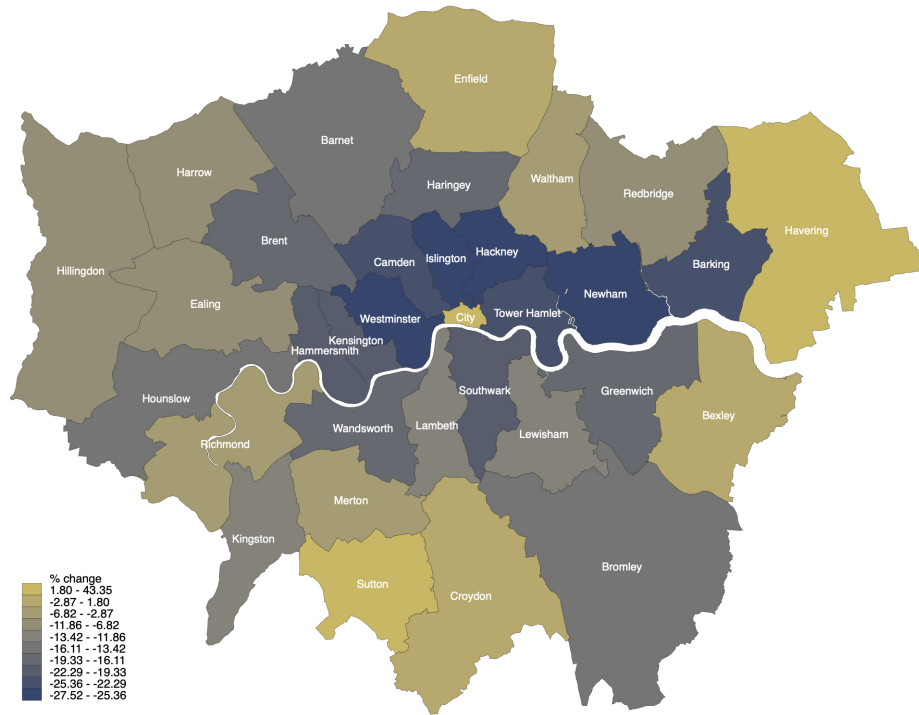


Figure 1: Map

This transition is complicated by disparities in spending cuts across councils, which disproportionately affected councils serving more deprived communities (Amin-Smith & Phillips, 2019). Intuitively, the effects of these reforms will not be felt equally in all councils, as not all councils are able to collect the same amount of tax per resident. The decrease in implicit redistribution by the government will have strong distributional consequences. This is shown in Figures (2) and (3), where the change in government budgets pre-reform did not depend on the level of deprivation. However, post-reform, we see a negative slope, suggesting the biggest cuts happened in the more deprived councils.

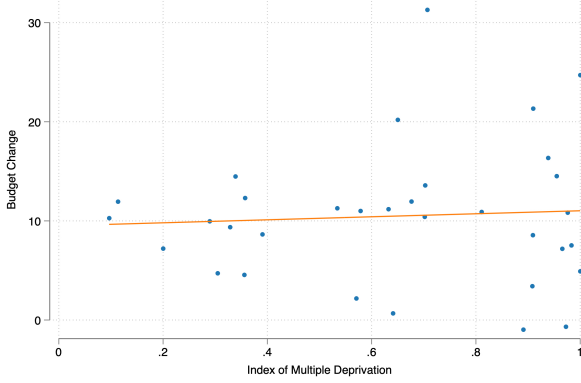


Figure 2: Budget changes 2007-10

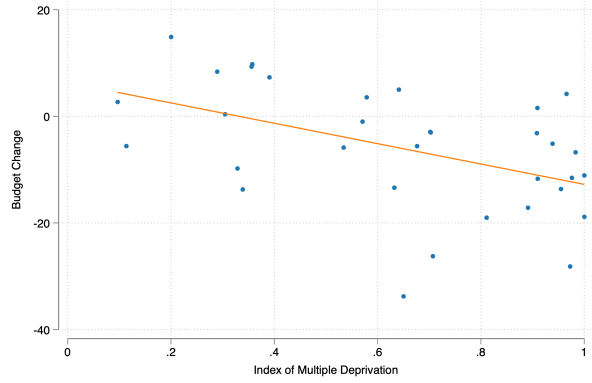


Figure 3: Budget changes 2011-14

Many inner London boroughs, burdened by large financial commitments due to having some of the most deprived areas in the UK, felt particular strain. In response to reduced funding, they have had to adapt by making significant cuts to more discretionary services, in order to provide relative protection for statutory and more acute services, such as education and social care. This had a particularly negative impact on the provision of services such as waste collection, housing, transport, and cultural services (Amin-Smith & Phillips, 2019).

As of today, the government intends to increase decentralisation of local government financing. The initial plan was to cut government grants further in 2021, while increasing business rate retention to 75%. This was put on hold due to the Covid pandemic, but is currently planned to be introduced in a future budget. The stated goal is to ultimately reach 100% business rate retention, while cutting most non-statutory grants to local authorities. These further policy changes will form the basis of my counterfactual, where I am to shed light on their effects in general equilibrium. As in any city, such a policy change will have both direct and indirect distributional consequences. I will show how workers respond, in particular in terms of residential location choice, and how this will ultimately spillover to equilibrium prices and wages.

2 Model

In this section I will explain the components of the quantitative urban model, which builds on recent advancements in the literature Ahlfeldt et al. (2015). I will consider a generic city with a number N of discrete locations indexed by $r, l \in S = 1, 2, \dots, N$ where r denotes the location of residence and l the location of work.

Each location is defined by its borders that endow it with a fixed amount of land, K_r . These locations are all connected and there is a given commute time between them. Workers and firms can optimally choose where to locate, with each location differing in residential amenities and productivity. In the rest of this section, I will detail each element of the model and the relevant derivations.

Workers

The city is embedded in a wider economy, and is populated by L workers. The workers are **homogenous**, supply one unit of labour, and consume housing and a freely tradable final good. The wider economy offers an outside option for all workers at a constant level of utility Υ . In addition, each worker receives an independent idiosyncratic taste shock for each residence-workplace pair.

The worker will thus optimize their location choices (residential and workplace) based on wages, house prices, commute costs, and the idiosyncratic taste shock. In line with the literature, I assume their preferences are Cobb-Douglas, and can therefore represent their utility function as:

$$U_{rl}(\omega) = \frac{z_{rl}(\omega)B_r}{d_{rl}} \left(\frac{C_{rl}(\omega)}{\alpha} \right)^\alpha \left(\frac{H_{rl}(\omega)}{1-\alpha} \right)^{1-\alpha} \quad (1)$$

The utility function $U_{rl}(\omega)$ represents the utility of a resident-worker from location r working at location l . The utility function has three parts: (1) residential amenities, $z_{rl}(\omega)B_r$, (2) consumption of goods, $(C_{rl}(\omega)/\alpha)^\alpha$, and (3) housing consumption, $(H_{rl}(\omega)/(1-\alpha))^{1-\alpha}$, all divided by the commuting cost. The parameter α denotes the share of income spent on consumption of goods, while $z_{rl}(\omega)$ is the idiosyncratic shock and $d_{rl} = (t_{ni})^\delta$ is the commuting cost, which is a function of travel time t . The idiosyncratic shocks are assumed to be independent draws from a Fréchet distribution with cdf:

$$\text{Prob}[z_{rl}(\omega) \leq z] = e^{-z^{-\varepsilon}}$$

This is a simplified distribution with a single shape parameter and no location parameters, which means that the draw probabilities are identical across locations. The shape parameter ε governs the variance of the draws.

An important change with respect to previous models in the literature is in the residential amenity term B_r . This is the product of the fundamental amenity, b_r , and the level of government spending on local public goods elevated to the relative importance of public goods, G_r^η . In the baseline model, G_r is normalised to 1 such that the fundamental amenity term can be backed out in the absence of policy intervention. **This essentially assumes that local public goods were equally distributed across the city prior to the policy change.** The parameter η governs the relative importance of local public goods with respect to the fundamental amenity.

$$B_r = b_r G_r^\eta \quad (2)$$

I assume no externalities on the residential amenity term in this model. Similarly, there are no spillovers. This is because the literature has shown spillovers to decay fast with distance (Ahlfeldt et al., 2015), and in my specific setting the boroughs are very large and would thus internalise all local externalities. By maximizing the utility function, we obtain the following relationships:

$$P_r C_{rl}(\omega) = \alpha w_l, \quad Q_r H_{rl}(\omega) = (1-\alpha)w_l \quad (3)$$

where w_l is the wage level at workplace location l . These two conditions equate the income spent on each consumption category, αw_l and $(1 - \alpha)w_l$, to the respective expenditure on goods ($P_r C_{rl}(\omega)$) and housing ($Q_r H_{rl}(\omega)$). This leads to the indirect utility function:

$$V_{rl}(\omega) = \frac{z_{rl}(\omega) B_r}{d_{rl}} \frac{w_l}{P_r^\alpha Q_r^{1-\alpha}} \quad (4)$$

One can clearly see that this depends positively on the idiosyncratic shock, wages, and amenities, and negatively on commuting costs and prices (housing and goods). Workers will first observe the realisation of their idiosyncratic shock, and then will make a residence-workplace pair decision optimally according to their indirect utility. Thanks to the properties of the Fréchet distribution, this will result in commuting probabilities:

$$\lambda_{rl} = \frac{(B_r w_l)^\varepsilon (d_{rl} P_r^\alpha Q_r^{1-\alpha})^{-\varepsilon}}{\sum_{k \in \mathcal{S}} \sum_{j \in \mathcal{S}} (B_k w_j)^\varepsilon (d_{kj} P_k^\alpha Q_k^{1-\alpha})^{-\varepsilon}}. \quad (5)$$

This expression is not surprising and just re-iterates what we know from the indirect utility function. The probability of any given worker choosing a certain commute increases with the wage and the residential amenity, and decreases with commuting costs and prices. This however is held in comparison to all the other possible combinations within a city, and thus we get an expression for all the commuting probabilities. From the unconditional probabilities above, we can also easily derive the conditional commuting probabilities:

$$\lambda_{rl|r} = \frac{(w_l/d_{rl})^\varepsilon}{\sum_{j \in \mathcal{S}} (w_j/d_{rj})^\varepsilon}. \quad (6)$$

Differently to the unconditional probabilities, these only depend on wages earned and on the commute costs. They are therefore independent of the other residential characteristics (amenities and prices), as they are conditional on residing in location r .

The commuting probabilities allow us to determine both the residential and the worker population for each location. This is done by multiplying the unconditional probabilities by the total population and adding them up. They are summed by residential location to determine the total number of workers, and by workplace location to determine the total number of residents.

$$R_r = \sum_{l \in \mathcal{S}} \lambda_{rl} L \quad (7)$$

$$L_l = \sum_{r \in \mathcal{S}} \lambda_{rl} L \quad (8)$$

Labour Supply

Open city

Under the assumption of an open city, labour supply to the city is a simple constant elasticity function that depends on the attractiveness of the city relative to the wider economy (reservation level of utility, Υ). The elasticity of labour supply to the city is denoted by σ .

$$L = \left(\frac{\bar{U}}{\bar{\Upsilon}} \right)^\sigma \bar{L}, \sigma > 1 \quad (9)$$

where $\bar{U}$ is the expected utility of moving to the city of a worker. The fraction of total labour $\bar{L}$ that moves to the city and creates the city's population L thus depends on the elasticity of labour supply parameter σ . From our previous derivations of utility and of commuting probabilities, we can calculate the expected utility of the city as:

$$\bar{U} = \bar{\gamma} \left(\sum_{n \in \mathcal{S}} \sum_{i \in \mathcal{S}} (B_n w_i)^\varepsilon (d_{ni} P_n^\alpha Q_n^{1-a})^{-\varepsilon} \right)^{1/\varepsilon} \quad (10)$$

where $\bar{\gamma}$ is the constant value of the Gamma function. It is worth noting that in equilibrium, the ex-ante expected average utility will be equal to the reservation level of utility.

Closed city

In the model that I will be estimating in this paper, I will be adopting a closed city framework. Therefore, there will be no choice to move in or out of the city, and the population consists of a fixed number L of workers. Generally, most urban models embed the city in a wider economy that allow the labour supply to adjust such that average utility is equalised to a constant level of utility as an outside option. I will abstract from this in the main model, as it simplifies the estimation and would not strongly affect results. Labour supply to the city is therefore taken as given and is fixed.

Firms

There are perfectly competitive firms distributed across the city, producing a final good using labour and floor space. The production function is a standard Cobb-Douglas:

$$Y_l = A_l (L_l)^\beta (H_l^B)^{1-\beta}, \quad \beta \in (0, 1), \quad (11)$$

where for each location l , A_l is the productivity, L_l is the labour input, and H_l^B is the floor space. The productivity term is given by:

$$A_l = a_l \left(\frac{L_l}{K_l} \right)^\chi, \quad (12)$$

I am assuming that productivity depends on a fundamental factor and an externality (density), but once again I am not including spillovers for the reasons previously explained. As with residential externalities, the fundamental productivity a_l is exogenous, while the density externality depends on L_l , K_l , and χ , where χ is the elasticity of productivity to employment density. This is the crucial parameter that determines the strength of agglomeration.

It is worth noting that there is no effect of government spending on productivity. I am assuming that government spending contributes to residential amenities only and not productivity, since the type of services they provide are geared towards residents (schools, garbage collection, health and social services, etc).

This is a perfectly competitive setup. Firms optimise in the face of free-entry and zero profits, which leads to the following expression for the equilibrium relationship between input and final good prices.

$$P_l = \frac{w_l^\beta q_l^{1-\beta}}{A_l \beta^\beta (1-\beta)^{1-\beta}} \quad (13)$$

where w_l denotes wages, P_l denotes the price of the final good, and q_l denotes floor space prices for commercial use. As the final good is freely tradeable, I normalise prices $P_l = 1$. The first order conditions of the cost minimisation problem also lead to the following relationships:

$$w_l L_l = \beta Y_l, \quad q_l H_l^B = (1-\beta) Y_l \quad (14)$$

Construction

The city has a perfectly competitive construction sector that develops available land. As previously explained, each location is endowed with a fixed amount of land K_l , of which T_l units of land are available for development (excludes parks, monuments, canals, etc). The construction sector provides floor space by combining land with a local productivity shifter. As will be shown below, this is analogous to a Cobb-Douglas production function where some form of capital is combined with land, in line with Epple et al. (2010). Floor space is then allocated to either the residential sector or the commercial sector. For simplicity, the model does not distinguish between these two types of floor space - they are perfectly interchangeable and thus the price of residential floor space will equal the price of commercial floor space ($Q_l = q_l$).

Construction firms will therefore have a production function of floor space specified as:

$$h_l = \kappa_l T_l^\nu \quad (15)$$

In this context, κ is a location specific cost shifter, and can be interpreted as some location specific cost $X^{1-\nu}$. The parameter ν can therefore be interpreted as the share of land in construction costs or alternatively the elasticity of construction density to land availability. In the competitive equilibrium, constructors aim to maximise profits $\pi_l = Q_l h_l - r_l T_l$, where r represents land rents and the only input cost to construction (the rest is absorbed by the location specific shifter). Given that construction is perfectly competitive, profit maximisation will lead to:

$$r = \pi_l^* = Q_l \nu \kappa_l T_l^{\nu-1} \quad (16)$$

We can then calculate the total rents received by landowners as $F_l = r_l T_l$.

Market clearing

To bring the model together in equilibrium, we need two market clearing conditions. The first is relative to workers, and it states that the number of workers in a given location l is equal to the total number of workers commuting to that location. It can be formalised by equating the number of workers of a given location to the sum, for all residential locations,

of the conditional probability of commuting to that location multiplied by its residential population:

$$L_l = \sum_{r \in \mathcal{S}} \lambda_{rl|r} R_r \quad (17)$$

The next market clearing condition dictates that floor space demand must equal floor space supply in equilibrium for all locations:

$$\frac{(1 - \alpha) \bar{w}_l R_l}{Q_l} + \frac{(1 - \beta) w_l L_l}{\beta Q_l} = h_l \quad (18)$$

The first component on the left hand side is the demand for floor space from residents, this can be backed out from worker utility maximisation. The second component is the demand for floor space from firms, which can be backed out from the first order conditions of the firm problem, using the conditions in (14). As there is no distinction in how floor space is used for either case, the sum of these two demands must equal the total supply in each location h_l .

Equilibrium

Given the above, I can define the equilibrium of the city. The equilibrium definition is in line with similar models from the literature (Sturm et al., 2023), whose proofs of existence and uniqueness, as well as inversion, will also hold for this model.

City Equilibrium *Given the set of structural parameters of the city $(\alpha, \beta, \eta, \varepsilon, \delta, \nu, \chi, \sigma)$, a level of utility in the wider economy (Υ) , the constant $(\bar{L})$ in the labour supply function (9), the normalised level of government spending (2), the constants (κ_l) in the floor space supply equation (15), and fundamentals in each location of the city (a, b, T) , and a matrix of travel times between each location in the city (t) , the general equilibrium of a city is characterized by the total city population (L) and vectors of wages (w) , land rents (r) , floor space prices (Q) residential populations (R) , employment at each workplace (L) , total floor space (h) that are determined by a system of seven equations: utility maximization and optimal residential choice of by workers (7); commuter market clearing (17); zero profit condition in production (13); supply of floor space (15); zero profits of developers (16); floor space market clearing (18) and labour supply to the city (9).*

3 Empirics

In this section I will describe all the empirical aspects of the paper. This will start with a preliminary description of the calibration procedure in section 3.1, including a discussion on how each parameter will be retrieved. Section 3.2 will describe the data that will be used. This will be followed by section 3.3 that will go through the reduced form components of the estimation. Finally, in section 3.4 I will explain the estimation procedure of the structural model.

3.1 Calibration

In this first part of the section, I will describe all the parameters in the model and how I will approach them. There are 8 structural parameters in the model, as shown in Table 1. For the purposes of my estimation, I will only need 7 parameters as I am maintaining a closed city, and therefore have no need for σ .

Table 1: Structural Parameters of the Model

Parameter	Description	Calibration Approach
α	Worker expenditure share on consumption	Data
β	Firm expenditure share on labour	Data
σ	Elasticity of labour supply to the city	n/a
δ	Elasticity of commuting cost wrt travel time	Estimation
ε	Fréchet shape parameter	Estimation
η	Public goods relative importance	Estimation
ν	Share of land in construction costs	Estimation
χ	Elasticity of productivity wrt density	Estimation

The first set of parameters can be directly informed by available data. The share of worker income spent on housing ($1 - \alpha$) is set at 0.4, according to London data from the ONS (2022b). Therefore, α is set to 0.6. The share of firm expenditure on labour (β) is set equal to the labour share of income, which is 0.6 in the UK according to the ONS (2022a).

The next set of parameters are those relating to worker supply. These are composed of $\delta, \varepsilon, \sigma$. As previously explained, the elasticity of labour supply σ drops out of the model. We are therefore left with the commuting costs with respect to travel time δ and the Fréchet shape parameter ε . The product of these two will be estimated using a gravity regression, as will be explained in section 3.3.2. Using evidence on the wage distribution I will be able to recover ε , which will then enable me to back out δ .

The next structural parameter to consider is the relative importance of public goods in the amenity equation. This is represented by η , and I will estimate it by conducting a boundary discontinuity Diff-in-Diff analysis. The details are set out in section 3.3.1.

The final two parameters, ν and χ will be estimated via the structural model, as explained in section 3.4. The first represents the share of land in construction costs. This parameter is crucial in curtailing agglomeration forces, and leads to a balance between density and sprawl. I will later show how it is identified and estimated. The second parameter, χ , represents the elasticity of productivity to density of economic activity. This too will be identified and estimated via the model, and is the central agglomeration force in this model.

3.2 Data

In this section I will describe all the data used in the estimation of the model. All data sources have either been procured directly from publicly available datasets or have been derived using publicly available sources. I have compiled data for some specific years to take advantage of the local funding change, as will be explained when relevant. The data is compiled at the borough level, as that is the geographic unit of analysis for the model.

Population: The first set of data refers to residential and employment population. This is the number of workers working and residing in each borough. The residential data was provided by the GLA Intelligent Unit (2018). It contains detailed resident statistics for both the ward and borough level. Similarly, the employment data is also supplied by the GLA through its GLA Economics research team (Hope, 2017), which provides detailed estimates for the number of workers by job type and industry in each borough. Both of these data are on a yearly basis.

The model’s structure requires that the number of workers equals the number of residents across the city. To account for this fact, the residential data is scaled down to equal the total number of employed workers in the GLA data. This has the implicit assumption that the employment rate is the same across boroughs, and also assumes that no workers are crossing the city boundaries in either direction. Given the low proportion of commuters between the Greater London area and the rest of England, this seems reasonable.

Area, land use The analysis requires detailed area data on Greater London. Specifically, land area data of London, by borough, is procured directly from the GLA which distributes Ordnance Survey data from the ONS (2020). This includes detailed area measures, shape-files, and location of postcodes which will be used in the boundary discontinuity analysis. In addition, further data on land use by borough will be used in the analysis. This provides information on the area of utilised land, and allows the exclusion of green space, paths, rail, road, and water. The data is maintained by the GLA but distributed by what was then known as the Department for Communities and Local Government (2005).

Wages: Estimates of average wages by borough will be required to estimate the Fréchet parameter ε . Specifically, the data required is average wage by borough of employment, and not residence. This was once again obtained from the ONS through a partial release of ASHE data (2022c). From the same data, I can also extract average daily flow of commuters between boroughs.

Local Government Spending: The next important aspect of data necessary is information on local public finances. The Department for Levelling Up, Housing and Communities provides detailed financial records for all local authorities for the past twenty years. From these records, I have been able to reconstruct the budgets available by London Borough for the years from 2007 to 2019, which offer enough bandwidth around the policy change of 2010/11.

House Prices: This aspect of the data was crucial for the border discontinuity analysis, in addition to assessing the fit of the model. I use data provided by Chi et al. (2021), which contains details on all house sale transactions in the UK since 1994 (in the updated dataset). This comes together with information on the price of sale and characteristics of the house, such as the post-code, decade of construction, EPC rating, and other information.

Commuting Time: A pivotal aspect of this study is the data on commuting times, which I calculated from the central point or centroid of each borough. The API for Travel for London is employed for these computations, with a focus on public transport as the mode of travel. Given the shape and size of different boroughs, the matter of intra-borough travel time is more complex. I have two different values: the default is standardizing it to 15 minutes for all boroughs, while the alternative is a function of the average width of boroughs. This choice does not affect my results.

3.3 Reduced Form

In this section I will describe two stand-alone analyses that are used to calibrate parameters in the model. The first one is a border discontinuity DiD regression, which is used to estimate the η parameter. The second will be a standard gravity regression. I have labelled these as reduced form as I temporarily abstract from the structure of the model.

3.3.1 Border Discontinuity Diff-in-Diff

One of the defining features of this model is the introduction of local public goods in the amenity term. As shown in equation (2), this consists of the introduction of a normalised value of local public spending. The term is multiplicative and elevated to η , which signifies the relative importance of local public spending with respect to the fundamental amenity value of the location. The estimation of the parameter η is therefore critical.

To estimate the parameter, I will take advantage of the many border discontinuities between boroughs in London. The concept behind the identification strategy is simple: if we take a group of houses on either side of the border, the only thing that could differentiate their value is the borough they belong to. This is because they are spatially very close to each other, would generally have the same fundamental amenity value (in terms of local beauty, services, restaurants), and would have the same commuting times to different work locations. Therefore, the identifying assumption is that the only difference between nearby houses on either side of the border are the excludable services provided by councils, which will be captured by the level of council spending G .

The idea of applying a simple regression discontinuity design at the boundary however has a few issues. The main is the direction of causality: higher spend areas can generally spend more if their real estate is more valuable, as they have a larger tax base. Therefore, I will be doing a border discontinuity difference in difference design, where I will leverage the large changes in council funding in 2010/11 to determine the effect on the change in house prices over the same period. Given that η is an exponent, the regression will have a log specification

to capture η as a coefficient. I will also specify the regression in first differences as I have two periods and a continuous “treatment”, similar to Card (1992):

$$\Delta \ln P_{ijr} = \alpha + \beta \Delta \ln G_r + X_j + \epsilon_i \quad (19)$$

where $\Delta \ln P_i$ is the difference in log average prices of postcode i in border j and borough r . X_j will capture fixed effects of the border and area - more on this below. To run the regression, I identify all post codes within a 500m threshold of the border, as shown in Figure 4. In the UK, and London in particular, post codes are very granular, allowing me to identify the location of the building up to some meters. This threshold is in line with the literature (Michaels et al., 2021; Black, 1999), but the results are robust to alternative thresholds. I have also introduced a 50m buffer around the threshold from which I am dropping all postcodes to minimise the effect of spillovers and possible cross-border measurement issues with post-codes. The specification is robust to modifying or removing the buffer.

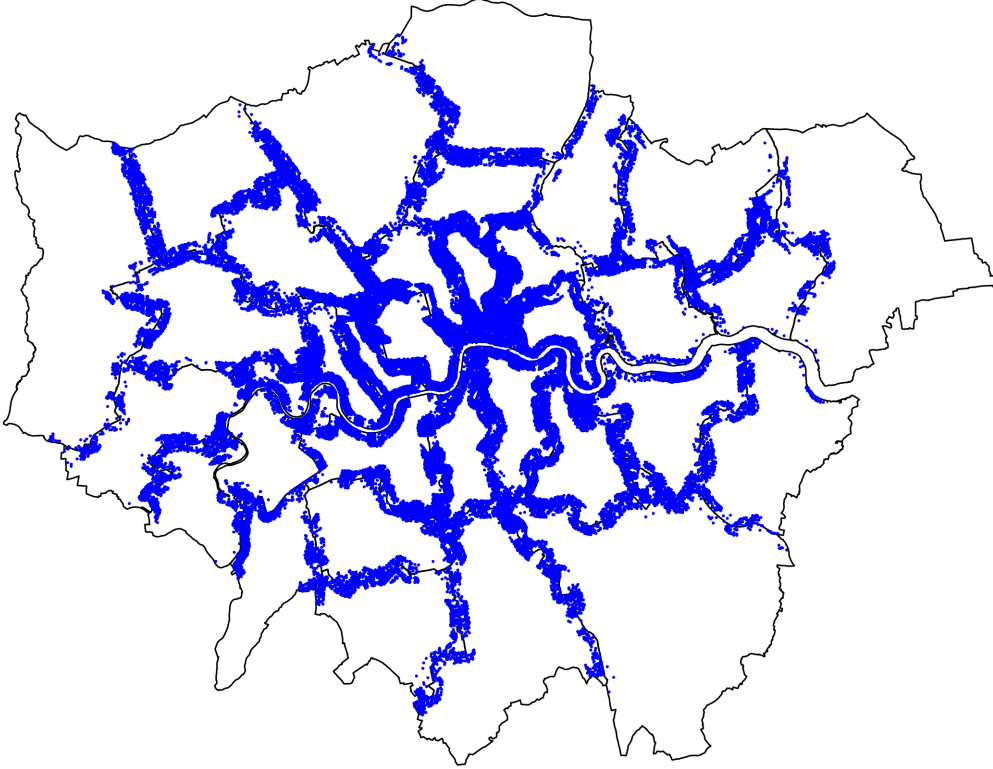


Figure 4: Post codes identified within 500m of borough boundary

After selecting the postcodes, I match all house sales in London to the relevant postcodes. I then create border dummies and border-segment dummies for every border (X_j). For clarity, there is a dummy to capture general effects around a specific border (e.g. Islington-Camden), in addition to a dummy to capture local fixed effects every one kilometre of border - the idea being that groups of neighbouring houses share the same local amenities. The results are robust to creating smaller border-segments. Finally, I will run the analysis with 2008-2010 data for the pre-period and with 2012-2014 data as the post-period, thus excluding the 2011

financial year which was the first year of implementation of the new budget rules. The results are presented in Table 2.

There are three different specifications. The first one only considers the border fixed effects, the second only considers border segment fixed effects, while the third includes both. My preferred specification is column (3), as it can best capture any local characteristics and therefore isolate the effect of government spending. This leads to an estimate of $\eta = 0.088$. To exemplify, this means that if the spending of a borough were to double, house prices in that borough would increase by 8.8%. This is once again in line with the literature, particularly regarding school districts (Black, 1999).

Table 2: Effect of Spend Change on Price Change, 2008-10 to 2012-2014

	(1)	(2)	(3)
Spend Change	0.0654** (0.029)	0.0884** (0.035)	0.0910** (0.036)
Border FE	Yes	No	Yes
1km Segment FE	No	Yes	Yes
Buffer	50m	50m	50m
Observations	18200	18200	18200

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The diff-in-diff analysis does have some threats to its identification. The first is that it requires an averaging of house prices by post-code, both pre and post. This is because we will not necessarily see the exact same house sale in both periods. The robustness of the analysis therefore relies on the assumption that house sales over the three year period cover similar types of houses within the same post code. In addition, it requires that houses are, on average, similar within the same border segments. **The assumption is that there are no policy or historical differences that would lead houses on one side of the border to be more expensive because they are larger, newer, or more luxurious.** Finally, the identifying assumption implies that there are no other non-spend policy differences between boroughs that could impact house prices. All of these threats have not been addressed in enough detail in this section as the main purpose was to calibrate the η parameter, but will be addressed in future iterations of this paper.

3.3.2 Gravity

In this section of the analysis, I estimate a gravity equation for commuting between London boroughs. As previously explained, I have data containing all location combinations of the 33 boroughs in London, including the borough of origin, the borough of destination, the commute time in minutes, and the number of daily commuters. The dependent variable of interest is the commuting probability, calculated as the ratio of the flow between boroughs

to the total flow of commuters.

The main explanatory variable is the travel time between boroughs, and I control for borough-specific fixed effects, both for origin and destination boroughs. The empirical specification of the gravity model can be written as:

$$\ln P_{ij} = \alpha + \theta \ln T_{ij} + \gamma_i + \delta_j + \epsilon_{ij} \quad (20)$$

where P_{ij} is the probability of commuting from borough i to borough j , T_{ij} is the travel time between boroughs i and j , and γ_i and δ_j are the origin and destination fixed effects, respectively. α is the intercept, θ is the coefficient of interest, and ϵ_{ij} is the error term. In relation to the model, the estimated coefficient can be interpreted as $\theta = \varepsilon * \delta$. This will be used in the estimation of the wages in the first step of the structural estimation in section 3.4, where the estimation of ε will allow the backing out of δ .

All the results are shown below in Table 3. The baseline estimator I will use is OLS with origin and destination fixed effects, shown in Column 1. For robustness and in line with the literature, I will also include a Poisson pseudo-maximum likelihood (PML) estimator in Column 2. This is commonly used to help with zero-flow pairs (Silva & Tenreyro, 2006), which is however not a problem in my specification as all boroughs have positive commuting flows.

While the Poisson ML regression is a common model for count data, it assumes that the mean and variance of the data are equal, which is known as equidispersion. However, count data often exhibit overdispersion, where the variance is greater than the mean. This can lead to underestimated standard errors and incorrect inference. I have therefore also included an estimate using the negative binomial regression in Column 3. This relaxes the equidispersion assumption by including an additional parameter to account for overdispersion.

Table 3: Gravity regression coefficients

	(1)	(2)	(3)
Log Travel Time	-4.399*** (0.139)	-3.457*** (0.108)	-4.776*** (0.126)
Estimator	OLS	Poisson PML	Negative Binomial
Origin FE	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes
Observations	1056	1056	1056
R2	0.864		

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The results are statistically significant and generally consistent, and are close to estimates that have been found for London in 1921 (Heblich et al., 2020) and in the United States

(Monte et al., 2018). In the structural estimation I will use the OLS estimates, however the use of either the Poisson PML or Negative Binomial estimates does not affect the results.

3.4 Structural Estimation

In this section I will outline the steps taken to estimate the model, which follow closely the recursive structure adopted in Sturm et al. (2023). The first step will be to estimate the wage distribution using the conditional commuting probabilities and the gravity equation. I will then compare this to the actual wage distribution to back out the Fréchet parameter ε , with which I can estimate δ and use these parameters to infer the final wage distribution. The next step entails using data on height and land use, combined with inferred floor space demand, to estimate the construction elasticity parameter ν . This allows to compute floor space prices, which can then be used for the model inversion to back out amenity and productivity terms. Finally, I can use the productivity term to estimate the remaining parameter in the model, χ .

3.4.1 Wages and ε

The first stage involves using the conditional commuting probabilities in equation (6) combined with the first market clearing condition in equation (17). This establishes the following relationship:

$$L_l = \sum_{r \in \mathcal{S}} \frac{(w_l/d_{rl})^\varepsilon}{\sum_{j \in \mathcal{S}} (w_j/d_{kj})^\varepsilon} R_r \quad (21)$$

While I can observe L_l and R_r , I do not have values for $d_{rl} = (t_{ni})^\delta$. I can therefore estimate a rougher version of the model where I have an unadjusted wage estimate $\tilde{w} = w^\varepsilon$, and take advantage of the gravity coefficient $\theta = \varepsilon * \delta = 4.4$ estimated in section 3.3.2:

$$L_l = \sum_{r \in \mathcal{S}} \frac{\tilde{w}_l/t_{rl}^\theta}{\sum_{j \in \mathcal{S}} \tilde{w}_j/t_{kj}^\theta} R_r \quad (22)$$

In this case, I have all the parameters necessary to solve for unadjusted wages $\tilde{w}$. I do this by solving for the $\tilde{w}$ that minimises the absolute distance between L_l and the right hand side.

I now need to estimate the Fréchet parameter ε , which I can do by comparing the variance of the unadjusted wage distribution with the variance I find in the data (Ahlfeldt et al., 2015). This is because ε governs the mapping between the two ($\tilde{w} = w^\varepsilon$), which can be interpreted as ε scaling the variance of log wages: $\sigma_{\ln \tilde{w}}^2 = (\frac{1}{\varepsilon})^2 \sigma_{\ln w}^2$. I can therefore estimate ε as the reciprocal of the OLS coefficient for the regression of the variance of unadjusted log wages on the variance of log wages in the data. I am thus able to back out a value of $\varepsilon = 3.53$, which is only slightly lower than what has been found in the literature (Ahlfeldt et al., 2015; Eaton & Kortum, 2002).

With the value of ε , I am able to back out δ from the gravity coefficient. This allows me to solve for wages in the same way using the relationship in equation (21). In Figure 5 I show the spatial distribution of estimated wages, while in Figure 6 the scatter plot of estimated

wages on observed wages. Higher wages seem to be concentrated in the center of the city, with average wages in the north and the west of the city. The south and east seems to have the lowest wages. On the scatter we can see that the fit of estimated wages with the data is very good, with a coefficient of 0.98.

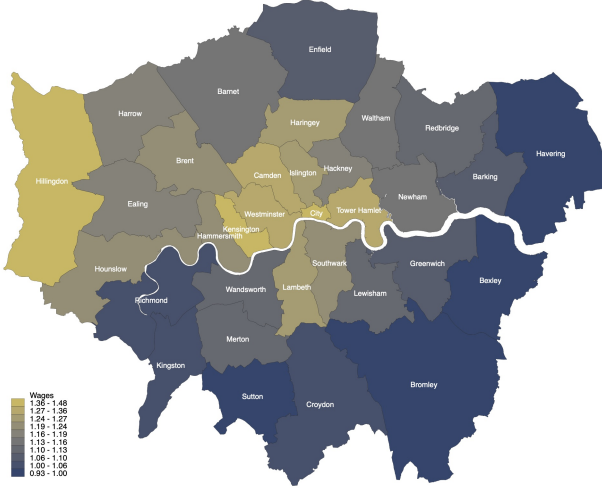


Figure 5: Wage Distribution

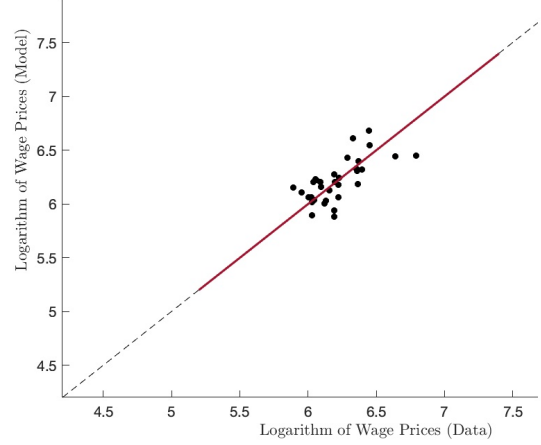


Figure 6: Wage Fit

3.4.2 Floor space prices and ν

The second stage of the estimation involves deriving the elasticity of construction costs to building height ν . Once ν is pinned down, the model can back out floor space prices which are equalised across residential and commercial markets by assumption. It is worth recalling that there is a fixed amount of land K_l , of which T_l units of land are available for development. To estimate ν , I combine the floor space supply function (??) with the equilibrium price of land from equation (16) to express height as:

$$h_i = \left(\frac{r_i}{\kappa_i(\nu - 1)} \right)^{1/\nu} \quad (23)$$

This can then be combined with the market clearing condition for floor space to obtain:

$$\nu \ln h_i = -\ln T_i - \ln \nu - \ln \kappa_i + \ln \left((1 - \alpha)\bar{w}_i R_i + \frac{\gamma}{\beta} w_i L_i \right) \quad (24)$$

The previous section of the estimation backed out wages, and the data have already informed γ, β and α . I have also already collected data on heights and land use, therefore the only remaining unknowns are ν and κ . In line with Sturm et al. (2023), I will treat κ as a structural residual in the estimation, and thus given any parameter ν , κ can be retrieved through the following moment condition:

$$\mathbb{E}[\mathcal{M}_h(X; \nu)] \equiv \mathbb{E}[\ln \kappa_i - \overline{\ln \kappa}] = 0 \quad (25)$$

where $\ln \kappa_i$ are the computed residuals and $\overline{\ln \kappa}$ their average. Essentially the moment restriction is used to minimise the deviation of the structural residual from the mean, mitigating the effect that variation in the structural residual can have on the height distribution. The parameter is then estimated via the following minimisation:

$$\hat{\nu} = \arg \min \left(\frac{1}{\sqrt{N}} \sum_{i=1}^N \mathcal{M}_h(X_i; \nu) \right)^\top \left(\frac{1}{\sqrt{N}} \sum_{i=1}^N \mathcal{M}_h(X_i; \nu) \right) \quad (26)$$

This leads to an estimated parameter of $\nu = 2.22$, which suggests a floor space supply elasticity of 0.82. This is quite low, however it is in line with estimates on housing supply elasticity in land constrained cities in the United States (Saiz, 2010). While London is not physically land constrained, construction is heavily constrained by regulation which could help explain the relatively low value.

Having backed out ν and κ , I can estimate floor space prices and rents from the land market clearing condition. In Figure 7 I show the spatial distribution of estimated rents, while in Figure 8 the scatter plot and fit of estimated floor space prices on observed prices. As expected, higher rents are concentrated in the center of the city, particularly in the City of London, Westminster and Islington. The periphery, especially in the East, has the lowest land rents. In terms of fit, there is still quite a bit of dispersion however it still fits very well with the data.

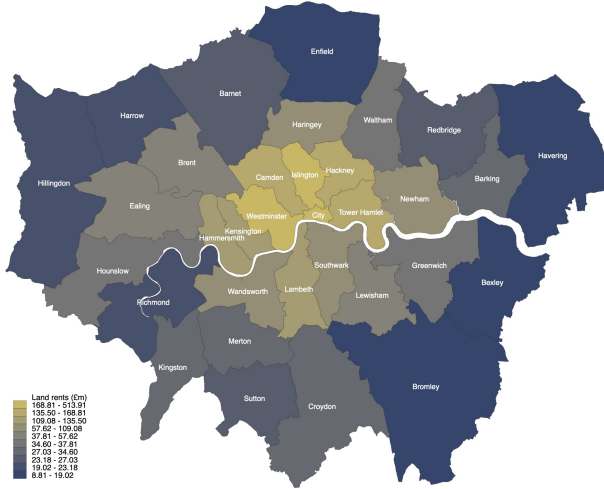


Figure 7: Rents Distribution

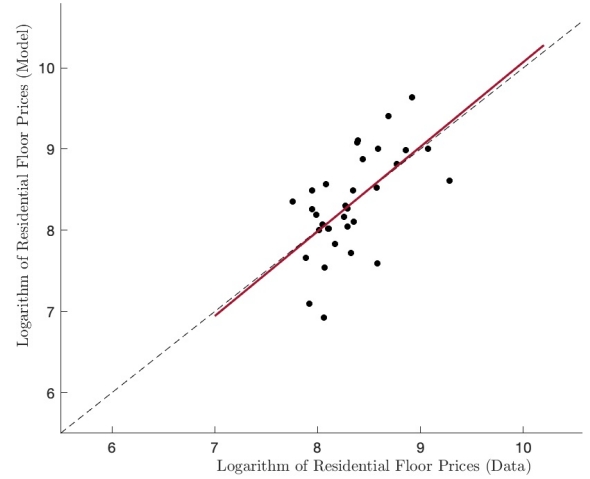


Figure 8: House Price Fit

3.4.3 Amenities, Productivity and Agglomeration

Given the estimates for wages and floor space it is now possible to retrieve estimates for productivity and amenities. Productivity can be backed out first by using the equilibrium relationship between input and output prices (13). Given that the price of the final good is

the numeraire (i.e. $P_l = 1$), and that the Cobb-Douglas coefficients are constant, I can pin down the level of productivity in each location up to a constant:

$$A_l \propto w_l^\beta q_l^\gamma \quad (27)$$

This simple relationship will thus reveal the productivity distribution across the city - locations paying higher wages and higher floor space prices must have higher productivity. Similarly, we can pin down the level of amenities up to a constant using the population equation (7) cobined with the commuting probabilities in (5). This leads to the following relationship:

$$B_r \propto \frac{R_r Q_r^{(1-\alpha)\varepsilon}}{\sum_{j \in \mathcal{S}} w_j^\varepsilon / d_{kj}^\varepsilon} \quad (28)$$

where $B_r = b_r$ as $G_r = 1$ in the baseline. Intuitively, this relationship tells us that locations with larger residential populations and floor space prices, combined with lower commuting market access to good work locations, must benefit from higher residential amenities. The results are mapped in Figures (9) and (10).

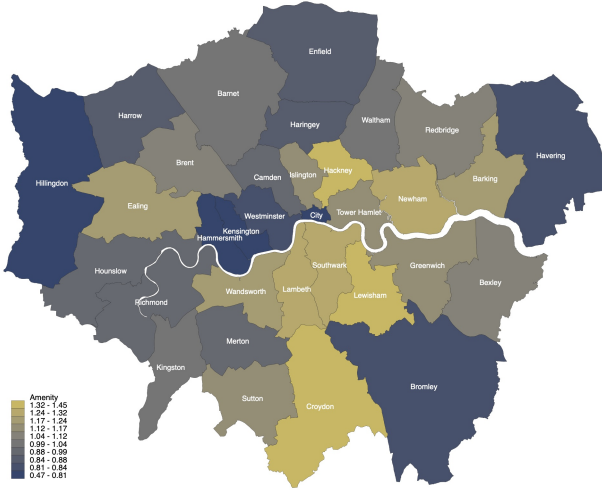


Figure 9: Amenities

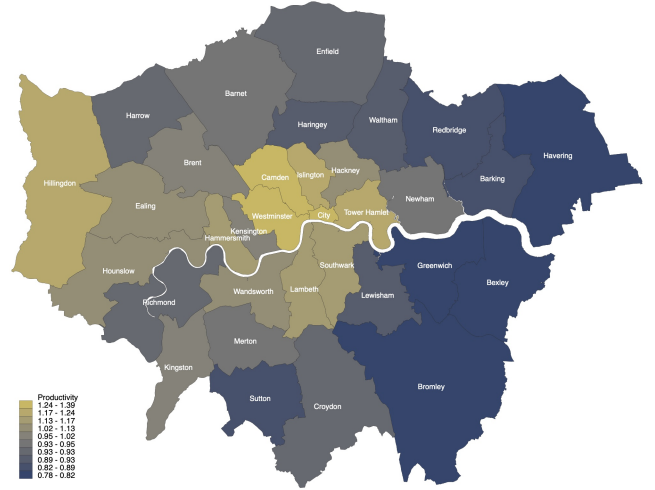


Figure 10: Productivity

Unsurprisingly, it is the central locations that benefit from the higher level of productivity, in line with the literature. Particularly the City of London, Westminster, and Camden. Productivity declines strongly in the eastern areas of London, while retaining an average level in western locations. The variation in productivity is quite high but in line with the literature, with the highest productivity areas having over 50% higher productivity than the lowest areas. An opposite pattern emerges with amenities, where it is the areas in the south and east that have the highest estimated amenities, while the most productive areas have among the lowest values. The variation in amenities is even stronger, with the highest areas having an 80% larger value than the lowest.

Finally, given the productivity level and the density, we can back out the density externality parameter χ . This will be done in line with the estimation method in Sturm et al. (2023), using a moment condition to pin down the fundamental productivity:

$$a_l = A_l \left(\frac{L_l}{K_l} \right)^{-\chi} \quad (29)$$

The moment condition will be specified similarly to the one used to estimate the construction elasticity parameter ν . The method will be to search for the best fitted value of χ to minimise the variation of fundamental productivity with respect to the average, essentially minimising the effect of the fundamental parameter. The moment condition will thus be specified as:

$$\mathbb{E}[\mathcal{M}_A(X; \chi)] \equiv \mathbb{E}[\ln a_l - \ln \bar{a}] = 0 \quad (30)$$

The parameter is then estimated via the following minimisation:

$$\hat{\chi} = \arg \min \left(\frac{1}{\sqrt{N}} \sum_{i=1}^N \mathcal{M}_A(X_i; \chi) \right)^\top \left(\frac{1}{\sqrt{N}} \sum_{i=1}^N \mathcal{M}_A(X_i; \chi) \right) \quad (31)$$

I estimate a value of $\chi = 0.1$. This is on the higher side with respect to the literature (Sturm et al., 2023; Ahlfeldt & Pietrostefani, 2019) but is in line with the estimated parameter in Bogota (Tsivanidis, 2019).

4 Counterfactual

I have now fully calibrated the model, and will use it to investigate the effect of the proposed new borough funding change. In particular, I will look at how prices, rents, wages, and population will change in each location given the policy change. The new policy, as explained in section 1, will further tie borough funding to local tax collection, increasing the disparity in funding between locations.

The first step of the counterfactual is to estimate the change in G_l . I have used data from the balance sheets of local councils, together with the proposed funding changes, to estimate the new level of government funding. This is expressed as a ratio over the previous level (in the model it was normalised to 1).

Given the change in G_l , and having estimated b_l and η , I can calculate the updated B_l . I can then use this in an iterative process, keeping all model parameters fixed, to back out the counterfactual estimates, which I will briefly summarise using the following steps:

1. Guess the new level of employment and residential populations in each location. Update the productivity term to account for the new employment density. As an initial guess I will use the values from the baseline.
2. Retrieve the new wages by using the conditional commuting probabilities and the commuter market clearing condition, as in section 3.4.1. This also allows to calculate the average wage by residential location.

3. Update the level of rents using the land market clearing condition and updated wages and populations. Use the new rents to compute the updated prices of floor space in each location.
4. Calculate the updated unconditional commuting probabilities from (5) using the updated amenities, wages, and floor space prices.
5. Use the new unconditional commuting probabilities to update residential and employment populations in equations (7) and (8).
6. Repeat all the steps replacing the residential and employment populations from step 5 back into step 1. Do this until the difference between the iterations of updated values converges to zero.

After carrying out the counterfactual simulation, I am able to back out all the new variables of the model. To put the effects into context, it is worth remembering that the funding cuts hit the central areas harder, barring the City of London, and largely benefited the periphery (Figure 1). We can see the effects in equilibrium on floor space prices and land rents:

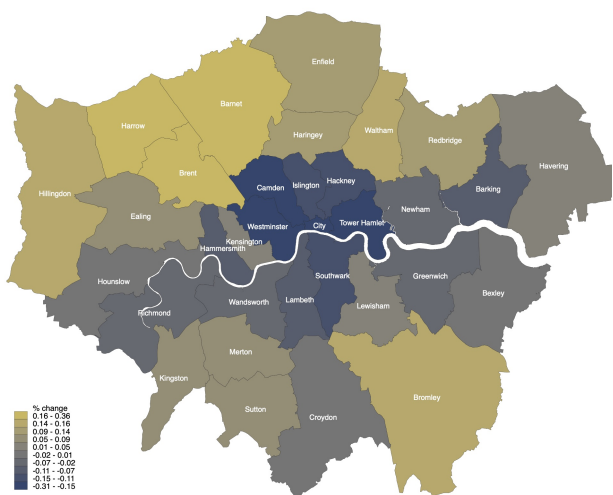


Figure 11: % Price Changes

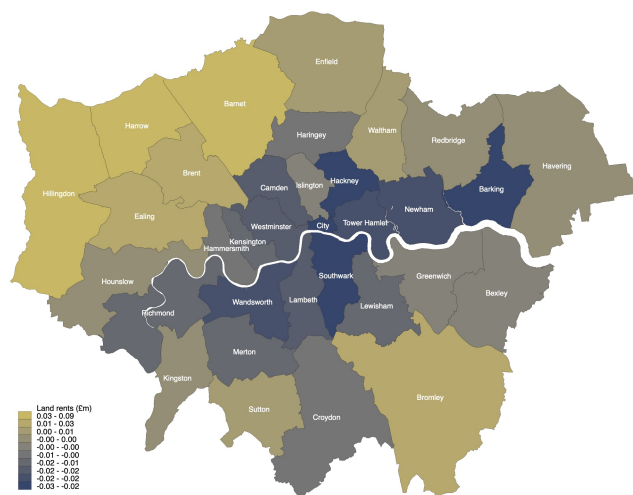


Figure 12: % Rent Changes

Interestingly, it seems that floor space prices and rents increase in periphery areas, which are not however the areas that received the biggest increase in amenity value. This could be explained by a few different factors, including that productivity is impacted by density, or that commuting patterns change enough to offset any increased amenity value. What remains true is that the areas that are hit with the larger cuts see a decline in floor space prices and rents. In addition, there is no big difference between rent change and floor space price changes, which is in line with the fact that the construction elasticity was relatively low, indicating it is hard for floor space to respond to prices.

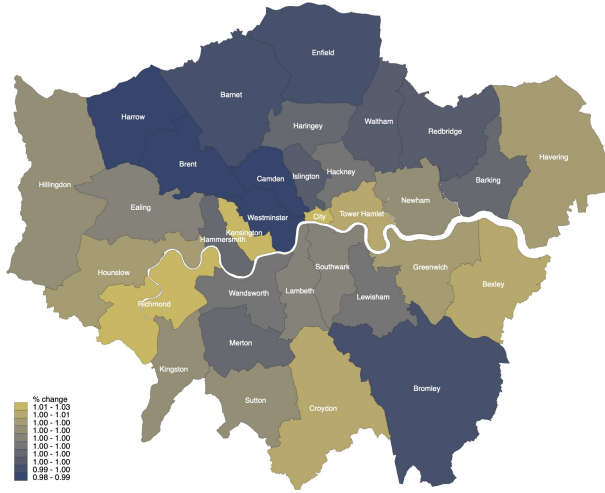


Figure 13: % Change Resident Space

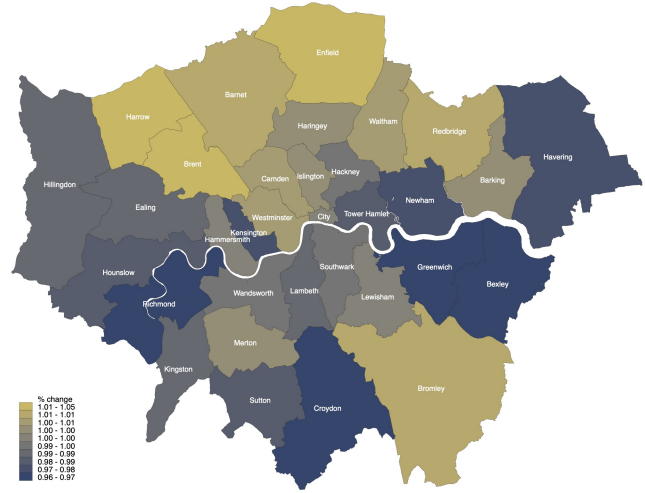


Figure 14: % Change Firm Space

To investigate further, I can look at the changes in the proportion of floor space used by residents (Figure 13) and businesses (Figure 14). As could be expected, it seems like there is a general increase in the proportion of floorspace going to residents in places that had the biggest increase in amenity value. The data also shows a big increase in firm activity in the northern side of the city in equilibrium. This suggests a relocation of economic activity, as certain areas have become relatively more attractive to live in.

Moreover, I can look at how wages by place of residence change, along with how the number of residents changes. Figure 15 shows that, in line with what one would expect, when amenities increase in the periphery through better services, the number of residents increases. Equivalently, the number of residents decreases in the city center. In addition, the wage of residents shows a stark increase in the north, similar to the change in rents. This is true even if not all areas see an increase in the proportion of firms, which can be explained from workers having a shorter commute from neighbouring boroughs.

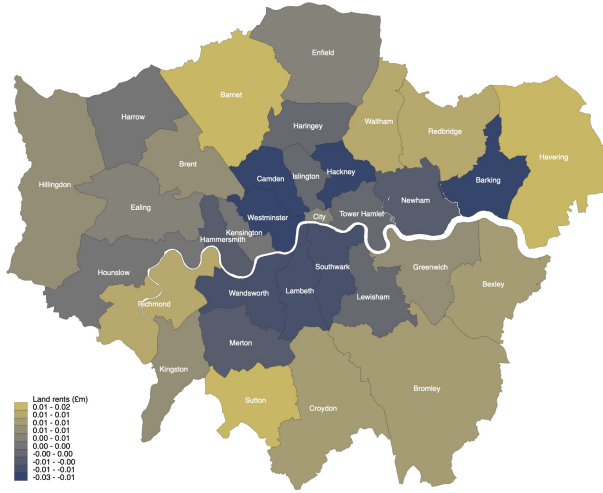


Figure 15: % Residents Change

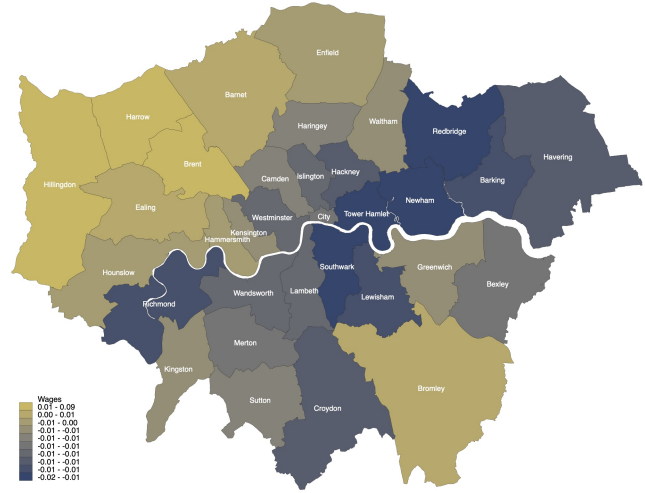


Figure 16: % Resident wage change

5 Conclusion and Discussion

In summary, this paper provides further evidence that underscores the viability of empirical urban models in exploring wage and price dynamics within city environments. By including local public goods in the analysis, I was able to implement a policy-relevant counterfactual that could potentially guide future policy development by shedding light on its effects and on those who would benefit or be disadvantaged by such reforms in equilibrium.

While the large geographic units of analysis limit what I can say about inequality within and across locations, it is clear that the policy does have distributional impacts. These occur directly, by increasing or decreasing the amenity value of the residential location, but also indirectly. There is re-sorting of both residents and businesses, which, given the transport network and commuting costs, leads to a new equilibrium wage level in each location. There are two key take-aways: public goods matter in urban economics models, and policy on local public goods can have strong distributional consequences.

Nonetheless, it is crucial to recognize that this paper is far from complete. There are several amendments and extensions necessary to enhance its robustness and the credibility of its findings. Future development of this paper will aim to augment the data, the model, and the counterfactual.

First of all, I would like to transition the analysis to a more granular geographic level such as wards, or potentially even more detailed. Due to constraints in time, I couldn't gather all the necessary data, estimate travel times, and calibrate the model for these smaller geographical units. This would certainly improve the validity of my estimates and therefore the power of the counterfactual.

With regards to the model, one enhancement I would like to make is the introduction of

heterogeneous workers. Ideally, there would be two groups of workers in the city, and the composition in each council is inherited due to historical patterns. In addition to mobile workers there would also be immobile workers, who currently receive benefits and would not be able to change borough of residence, as the benefits are provided by the borough. These workers take up a certain proportion of the council's budget, which therefore cannot be used to "attract" new residents and does not augment the amenity value. Due to a large degree of variation in their proportion across boroughs, this would represent another dimension of inequality that would be increased by the policy change, which I would like to capture in my model.

As for the counterfactual, there are a few avenues of expansion. One would be to allow labour supply to vary by setting the model as an open city. This would also allow me to capture the effects of the policy on labour supply. Another enhancement would be to make taxes an endogenous factor. Theoretically, business rates and council taxes should align with the value of floor space. This would mean that as floor prices increase in a certain location, so do taxes and local funding. It would be insightful to observe the consequences of such an adjustment in this context. In addition, I would also like to take the counterfactual further and say something about how inequality is affected through price segregation.

My hope is that these enhancements would serve to extend the study's relevance and potential applications, ultimately becoming a useful tool to assess the effects of local government funding policy in England and London more specifically.

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